

The Unemployment Rate

$$\text{Unemployment rate} = \frac{\text{Unemployed}}{\text{Labor force}} \times 100$$

- ▶ The denominator includes employed and unemployed people.
- ▶ In the miniature economy: $10/80 \times 100 = \mathbf{12.5\%}$.
- ▶ Dividing by the entire adult population would answer a different question.

Labor-Force Participation

The same economy

Employed	70
Unemployed	10
Outside the labor force	20
Labor force	80
Adult population	100

$$\text{Participation rate} = \frac{\text{Labor force}}{\text{Adult population}} \times 100$$

- ▶ This measure asks what share of the adult population is working or actively seeking work.
- ▶ In the miniature economy: $80/100 \times 100 = \mathbf{80\%}$.
- ▶ A decline can reflect retirement, schooling, caregiving, discouragement, or other reasons.

The Employment-Population Ratio

The same economy

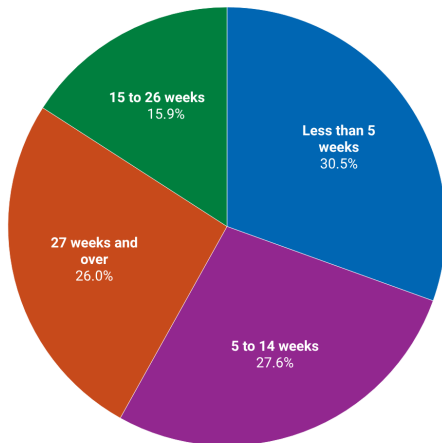
Employed	70
Unemployed	10
Outside the labor force	20
Labor force	80
Adult population	100

Employment-population ratio

$$= \frac{\text{Employed}}{\text{Adult population}} \times 100$$

- ▶ This measure does not depend on whether nonworkers are searching.
- ▶ In the miniature economy: $70/100 \times 100 = \mathbf{70\%}$.
- ▶ Use it with unemployment and participation to see whether a rate change reflects hiring or movement out of search.

Unemployment Duration, December 2025



Source: U.S. Bureau of Labor Statistics; chart © Worth Publishers, Figure 8.

Natural Rate of Unemployment

$$\text{Natural unemployment} = \text{Frictional} + \text{Structural}$$

- ▶ The natural rate is the normal rate around which actual unemployment fluctuates.
- ▶ It is not zero, permanent, or directly observed; economists estimate it.
- ▶ Labor-force composition, matching technology, institutions, and policy can change it.

$$\text{Cyclical component} = \text{Actual unemployment rate} - \text{Natural rate}$$

Changes in the Natural Rate of Unemployment

- ▶ **Changes in labor-force characteristics**

Examples: age mix; education and skills; share of new entrants.

- ▶ **Changes in labor-market institutions**

Examples: union coverage; temporary-work agencies; online job matching.

- ▶ **Changes in government policies**

Examples: job-training programs; unemployment benefits; minimum-wage rules.

Source: Chapter 8 publisher slides 25–29 and 34.

Reading the Population Counts

December 2025; thousands of people; not seasonally adjusted.

Measure	Published count
Civilian noninstitutional population, age 16+	274,816
Employed	163,720
Unemployed	7,003
Labor force	170,723
Outside the labor force	104,094

$163,720 + 7,003 = 170,723$: employed plus unemployed equals the labor force.

7,003 thousand means 7.003 million people. Totals may differ slightly due to rounding.

Source: BLS, December 2025 Employment Situation (January 9, 2026), Table A-1; core counts independently checked against FRED downloads.

Calculating the Unemployment Rate

December 2025

Thousands; unadjusted

Population	274,816
Employed	163,720
Unemployed	7,003
Labor force	170,723

$$u = \frac{U}{LF} \times 100$$

- ▶ **Choose the denominator:**
 $LF = 163,720 + 7,003 = 170,723$.
- ▶ **Divide:** $7,003/170,723 = 0.04102$.
- ▶ **Convert:** $u \approx 4.1\%$ of the labor force is unemployed.

Source: BLS, December 2025 Employment Situation (January 9, 2026), Table A-1. Both numerator and denominator are in thousands.

Calculating Participation

December 2025

Thousands; unadjusted

Population	274,816
Employed	163,720
Unemployed	7,003
Labor force	170,723

$$LFPR = \frac{LF}{P} \times 100$$

- ▶ **Choose the denominator:** the population $P = 274,816$.
- ▶ **Divide:** $170,723/274,816 = 0.62123$.
- ▶ **Convert:** $LFPR \approx \mathbf{62.1\%}$ of this population is in the labor force.

Source: BLS, December 2025 Employment Situation (January 9, 2026), Table A-1. The denominator excludes children under 16, active-duty military, and institutional residents.

Calculating the Employment-Population Ratio

December 2025

Thousands; unadjusted

Population	274,816
Employed	163,720
Unemployed	7,003
Labor force	170,723

$$EPOP = \frac{E}{P} \times 100$$

- ▶ **Choose the numerator:** employed people, $E = 163,720$.
- ▶ **Divide:** $163,720/274,816 = 0.59574$.
- ▶ **Convert:** $EPOP \approx \mathbf{59.6\%}$ of this population is employed.

Source: BLS, December 2025 Employment Situation (January 9, 2026), Table A-1. The unemployment rate and employment-population ratio use different denominators.

Optional Further Detail

The regular measurement tutorial is complete.

You have covered classification and the unemployment, participation, and employment-population rates.

The remaining slides are optional, for anyone who would like more detail:

- ▶ Broader measures, reasons, and duration of unemployment.
- ▶ The natural rate and an estimated cyclical gap.
- ▶ Extra spreadsheet practice and data-download links.

These additional methods are not required to complete the regular measurement tutorial.

Estimating the Natural Rate

In the chapter: natural unemployment = frictional + structural unemployment.

- ▶ **Frictional:** time spent finding a job. **Structural:** mismatches or labor-market institutions that prevent matches.
- ▶ **Who estimates it?** BLS measures actual unemployment. The noncyclical benchmark used here is estimated by **CBO**.
- ▶ **One CBO approach:** estimate normal unemployment for demographic groups, then weight their rates by their shares of the labor force.

The natural rate is not directly observed. BLS reason and duration tables do not separately count its two components.

Source: Chapter 8; CBO, Approaches to Estimating the Noncyclical Rate (2023); BLS, projections methods (2024).

Estimating Cyclical Unemployment

Estimated cyclical gap = Observed rate u – Noncyclical estimate u^*

Quarter	Observed u	CBO u^*	Gap (pp)
2019 Q4	3.60%	4.47%	-0.87
2020 Q2	13.00%	4.46%	+8.54

13.00 – 4.46 $\approx$ **8.54** percentage points above the estimated noncyclical rate.

The benchmark is estimated, not directly observed. A negative gap means unemployment is below that benchmark, not a negative number of unemployed people.

Source: BLS UNRATE; CBO NROU, via FRED. Display rounded; calculations retain full precision.

Downloading the Data

December 2025; United States; thousands of people; not seasonally adjusted.

Measure	FRED series	Download
Population	CNP16OV	CSV
Labor force	LNU01000000	CSV
Employed	LNU02000000	CSV
Unemployed	LNU03000000	CSV

Open each CSV in Excel or Sheets. Match the observation date before combining columns.

Source: BLS via FRED. Download the original BLS release PDF; Tables A-1, A-8, A-11, A-12, A-15, A-16.